

2Market: Customer Demographics and Marketing Channel Effectiveness

Business Context

2Market aims to better understand its **customer demographics, behaviour, and marketing channel performance**. The company seeks to identify how demographic factors influence purchasing patterns and which channels—such as social media, brochures, or bulk emails—generate the best sales results.

The overarching business objective is to **optimise marketing strategy** to increase overall sales and **improve return on investment (ROI)** from marketing expenditure.

To further refine the project scope, several key questions were raised for the board of directors:

- How will the company use the demographic and customer insights provided?
- Is there additional data available linking **purchased items to marketing channels**?
- What products, if any, are **targeted towards children**?

Analytical Approach

Data Cleaning and Preparation

The raw customer dataset was first cleaned using **SQL** and **Excel**. The data was imported into Excel to check for consistency and proper formatting. Columns were standardised (e.g., numeric and date formats), and a new variable, **age**, was created using the formula:

A search for duplicates through the **ID column** found none, and no missing values were detected.

During the data review, some irregularities appeared in the **marital status** column (entries such as *YOLO*, *alone*, and *absurd*).

- *YOLO* and *absurd* were removed, as they accounted for only 0.09% each of the total sample.
- *Alone* was reclassified as *single* for consistency.

Outliers were identified and removed from **income** (values exceeding 112,000) and **age** (values above 84 years). These extreme values likely represented data entry errors, as some cases exceeded 124 years of age.

Preliminary Analysis

Data was sorted and examined by **age**, **income**, and **marital status** to identify early relationships.

No significant correlation was observed between **age and income** or between **income and number of items purchased**. However, meaningful patterns emerged when analysing data by **country**, **education**, **marital status**, and **income**.

Cross-Referencing and SQL Analysis

To enhance understanding, data was cross-referenced using **country**, **marital status**, **age**, and **education** as key demographic features. This allowed the extraction of more detailed behavioural insights, particularly regarding households with children.

The cleaned dataset was then imported into **SQL**, where two new tables were created based on **customer ID**. These tables enabled analysis of **lead conversion rates** across marketing channels (social network, brochure, and bulk email).

Grouping data by **marital status**, **country**, and **education** revealed clear distinctions in the effectiveness of different marketing channels and customer response patterns.

Dashboard Design and Development

A dashboard was developed to visually communicate the findings from the analysis and highlight key demographic and behavioural insights.

- **Upper Section:** Focused on **revenue by marital status**, **customer distribution**, and **average income**, segmented by families **with** and **without children**.
- **Middle Section:** Displayed **customer education levels** as a percentage of the total population.
- **Lower Section:** Showed **lead conversion success rates** across different marketing media (social networks, brochures, and bulk emails).

Given the data richness, only one **interactive element** was included — allowing users to highlight a specific marital status to view its spending behaviour across the six best-selling product categories.

A **blue palette** was chosen for marital status and revenue charts to maintain consistency and a clear visual link between related insights. The **education chart** used shades of red, while the **marketing channel chart** featured neutral tones of black for clarity and accessibility.

Chart titles were kept simple and descriptive, ensuring immediate comprehension. The design prioritised **clarity**, **colour accessibility**, and **minimalism**, allowing users to quickly interpret key results despite the large data volume.

Patterns, Trends, and Insights

Customer Demographics

- The majority of customers are in their **40s to 60s**, with roughly half being **married or in a relationship**.
- The customer base is **highly educated** — about 50% hold a graduate degree, 20% a PhD, and 15% a master's degree.
 - This suggests a mature, well-informed consumer base likely earning higher incomes and exhibiting a greater focus on **quality over price**.

Geographic Distribution:

- Around **50%** of customers are from **Spain**, followed by **South Africa (15%)** and **Canada (10%)**.
- Other notable markets include **Australia, Germany, India, and the United States** (each approximately 5%).
- It may be valuable to explore whether 2Market stores carry products relevant to these national origins.

Product and Purchase Behaviour

- Approximately **half of all purchases** fall into the **liquor category**, followed by **meat (27%)**, then smaller proportions of **commodities, fish, chocolate, and vegetables**.
- Despite about **50% of customers having children**, there are **few products directly targeted at younger consumers**, except for chocolate — suggesting an untapped opportunity in the **family and child-oriented** product segment.

Marketing Channel Effectiveness

- **Digital channels** (social networks and bulk email) significantly outperform traditional media. Online conversions are **five times higher** than brochure-based conversions.
- The findings indicate that 2Market's marketing ROI can be maximised by **shifting investment toward digital channels** while potentially reducing spending on printed brochures.

Future analysis could explore **cost per conversion** for each channel (e.g., pay-per-click vs. brochure printing and distribution costs) and assess **which product types sell best through each channel** to further refine marketing strategies.

Conclusion

The analysis provides a comprehensive view of 2Market's customer demographics, purchasing behaviour, and marketing effectiveness.

Key findings include:

- A **mature, well-educated customer base** primarily from Spain, with notable international presence.
- **Liquor and meat** dominate sales, though there is **potential to expand child-focused product lines**.
- **Digital marketing channels** (especially social media and bulk email) deliver **significantly higher conversion rates** than traditional brochures.

These insights can guide 2Market in:

- Optimising **marketing spend allocation** toward high-performing channels.
- Developing **targeted campaigns** by demographic segment.
- Expanding product offerings to align with customer profiles and unmet needs.

Overall, this analysis supports 2Market's strategic objective of **improving sales performance and maximising ROI** through data-driven marketing and customer segmentation.